

Algorithm Lab

Week 4: Geometry Problems on $2^n \times 2^n$ Plane

Description

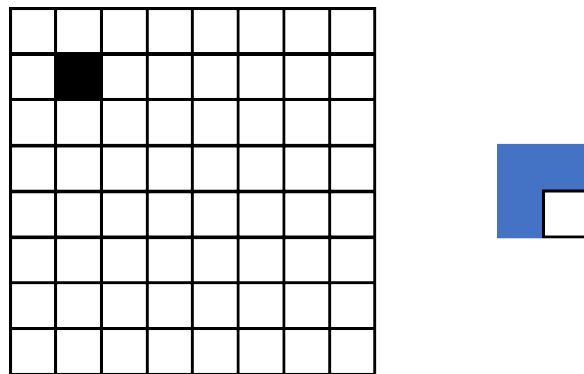
Divide and conquer is an algorithm design paradigm with followed steps:

1. If the answer is trivial, solve it.
2. If not, solve the problem recursively
 - 2.1. Divide the original problem to independent same-type subproblems.
 - 2.2. Solve subproblems.
 - 2.3. Aggregate the results of 2.2

For a given $2^n \times 2^n$ square, with $n \geq 1$, we always can divide it to 4 $2^{n-1} \times 2^{n-1}$ squares. Thus, if our problem is on a $2^n \times 2^n$ plane, we can consider design the solution with divide and conquer paradigm.

Tiling problem is a classical problem. Giving tiles and a field that you need to fill. That kind of problem asking for cover the whole filed with given tiles without overlaps. You can rotate the tile if needed.

L-shaped tile is a 2×2 tile with a clipped corner. L-shaped tiling problem is defined as followed:

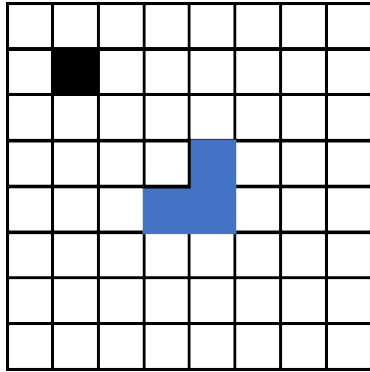


Instance: Square size $n \in \mathbb{N}$ and an obstacle $(x, y), 1 \leq x, y \leq 2^n$

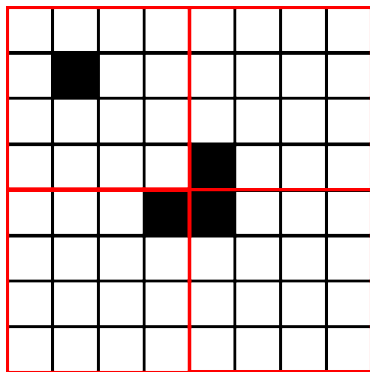
Result: Tiling layout with L-shaped tiles which covers all the normal cells.

Algorithm Design

- 1 If the target square is a L-shaped, finished.
- 2 If not, solve the problem recursively.
 - 2.1 Put one L-shaped tile on the center of the square
 - 2.2 Rotate the tile to let absented corner direct to the hole



- 2.3 Divide the problem to 4 subproblems



L-shaped tiling is just an example. In real-world, we solving many different problems on geometry system and image compression system by this kind of approach. For example, one day, you want to find out oil stations nearby you with Google Map, the app will find out your target as followed:

1. Get your current location
2. Decide the searching area (most of case, your screen)
3. If the area is small enough, list the search result.
4. If not, solve the problem recursively
 - 4.1. Divide the area to 4 areas
 - 4.2. Search targets on the 4 areas.
 - 4.3. Combine all the results

The skill also applied on many different things such like mobile network load balancing (3G, 4G, 5G), 2-D wavelet compression (JPEG2000), Human-flow counting with satellite, ..., etc.

Questions

1. How to applied above Google Map searching algorithm on a not square area?
2. Suppose the above function will return targets ordered by distance to the user. Please design a linear time other to aggregate the results from 4 sub-problems.
3. This skill is very useful on maintain a sparse matrix, please
 - 3.1. Design a sparse matrix structure with above skill
 - 3.2. Analysis time complexity in worst case on get-operation set-operation
4. Solving <http://oj.csie.ndhu.edu.tw/problem/ALG03B> and analysis your code.